Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: TP08

Subject: **Operating System**

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1. Create script(s) as following:
    a. Script “eonea” which prints the defined var. “x” with value 10.

```
sokmeak@sokmeaksaren:~/bin/Others$ cat eonea
#!/bin/bash
x=10
echo "The value of a : $x"
```

    b. Script “eoneb” which prints the defined var. “y” with value “Rani”.

```
sokmeak@sokmeaksaren:~/bin/Others$ cat eoneb
#!/bin/bash

y="Rani"

echo "The value of y $y"
```

    c. Run your scripts and take a screenshot.

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod +x eoneb
sokmeak@sokmeaksaren:~/bin/Others$ ./eoneb
The value of y Rani
sokmeak@sokmeaksaren:~/bin/Others$
```

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod +x eonea
sokmeak@sokmeaksaren:~/bin/Others$ ./eonea
The value of a : 10
sokmeak@sokmeaksaren:~/bin/Others$
```

2. Create script(s) as following:
    a. Script “etwoa” which prints the sum of two integers of var. “a” & “b”; e.g. 3 &5.

```
  GNU nano 7.2                                                    etwoa
#!/bin/bash
a=$1
b=$2

sum=$((a+b))
echo " $sum "
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./etwoa 3 5
 8
```

    b. Script “etwob” which prints the division of two integers of var. “a” & “b”; e.g. 20 & 5.

```
  GNU nano 7.2                                                    etwob
#!/bin/bash
result=`echo  " scale=6; $1 / $2 " | bc`
echo $result
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./etwob 4 3
1.333333
sokmeak@sokmeaksaren:~/bin/Others$
```

c. Script “etwoc” which prints the sum of two real number of var. “a” & “b”; e.g. 1.2 & 2.3.

```
  GNU nano 7.2                                                    etwoc
#!/bin/bash
result=`echo "scale=6;$1 + $2" | bc`
echo " $result "
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./etwoc 3.4 4.5
 7.9
```

d. Script “etwod” which prints the division of two real number of var. “a” & “b”; e.g. 2 & 3.

```
  GNU nano 7.2                                                    etwod
#!/bin/bash

result=`echo "scale=6;$1/$2" | bc`
echo " $result "
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./etwod 4.2 2.1
 2.000000
```

3. Create a command **mycal augment** which accepts any of the augments such as “2-4” or “4+2” or “7/8” or “3*7” and then take a screenshot of all cases.

```
  GNU nano 7.2                                          mycal.sh
#!/bin/bash

# 3. Command mycal augment
mycal() {
    if [ "$1" == "augment" ]; then
        echo "Performing calculation: $2"
        result=$(echo "$2" | bc 2>/dev/null)
        if [ $? -eq 0 ]; then
            echo "Result: $result"
        else
            echo "Invalid calculation. Please use valid expressions like 2+3, 5-2, 4*3, 10/2."
        fi
    else
        echo "Usage: mycal augment 'expression'"
    fi
}

# Call mycal anywhere
if [ "$0" == "./mycal.sh" ]; then
    mycal "$1" "$2"
fi
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./mycal.sh augment "4+4"
Performing calculation: 4+4
Result: 8
sokmeak@sokmeaksaren:~/bin/Others$ nano mycal.sh
sokmeak@sokmeaksaren:~/bin/Others$ ./mycal.sh augment "2-4"
Performing calculation: 2-4
Result: -2
sokmeak@sokmeaksaren:~/bin/Others$
```

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod  +x mycal.sh
sokmeak@sokmeaksaren:~/bin/Others$ ls
checkFileExtension  eoneb  etwob  etwod          getStarted      mycal.sh    user_management
eonea               etwoa  etwoc  fileOrganizer  helloworld.jar  organizeMe
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./mycal.sh augment "4/2"
Performing calculation: 4/2
Result: 2
sokmeak@sokmeaksaren:~/bin/Others$ ./mycal.sh augment "4*2"
Performing calculation: 4*2
Result: 8
```

4. Write a shell script that prompts the user for a name of a file or directory and reports if it is a regular file, a directory, or another type of file. (Using if else)

```
  GNU nano 7.2                                                    checker.sh
#!/bin/bash

read -p "Enter the name of a file or directory: " name
if [ -f "$name" ]; then
  echo "$name is a regular file."
elif [ -d "$name" ]; then
  echo "$name is a directory."
else
  echo "$name is another type of file."
fi
```

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod +x checker.sh
sokmeak@sokmeaksaren:~/bin/Others$ ls
checker.sh          eonea  etwoa  etwoc  fileOrganizer  helloworld.jar  organizeMe
checkFileExtension  eoneb  etwob  etwod  getStarted     mycal.sh        user_management
sokmeak@sokmeaksaren:~/bin/Others$ ./checker.sh
Enter the name of a file or directory: helloworld.jar
helloworld.jar is a regular file.
sokmeak@sokmeaksaren:~/bin/Others$ ./checker.sh
Enter the name of a file or directory: Others
Others is another type of file.
sokmeak@sokmeaksaren:~/bin/Others$ ls
checker.sh          eonea  etwoa  etwoc  fileOrganizer  helloworld.jar  organizeMe
checkFileExtension  eoneb  etwob  etwod  getStarted     mycal.sh        user_management
sokmeak@sokmeaksaren:~/bin/Others$ mkdir Test
sokmeak@sokmeaksaren:~/bin/Others$ ls
checker.sh          eonea  etwoa  etwoc  fileOrganizer  helloworld.jar  organizeMe  user_management
checkFileExtension  eoneb  etwob  etwod  getStarted     mycal.sh        Test
sokmeak@sokmeaksaren:~/bin/Others$ ./checker.sh
Enter the name of a file or directory: Test
Test is a directory.
```

5. Modify the previous script to that it accepts the file or directory name as an argument instead of prompting the user to enter it.

```
  GNU nano 7.2                                          superChecker.sh
#!/bin/bash

if [ $# -eq 0 ]; then
  echo "Usage: $0 <file_or_directory_name>"
  exit 1
fi
#get name of the user input
name=$1
if [ -f "$name" ]; then
  echo "$name is a regular file."
elif [ -d "$name" ]; then
  echo "$name is a directory."
else
  echo "$name is another type of file."
fi
```

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod +x superChecker.sh
```

```
sokmeak@sokmeaksaren:~/bin/Others$ nano superChecker.sh
sokmeak@sokmeaksaren:~/bin/Others$ ./superChecker.sh mycal
mycal is another type of file.
sokmeak@sokmeaksaren:~/bin/Others$ ./superChecker.sh mycal.sh
mycal.sh is a regular file.
sokmeak@sokmeaksaren:~/bin/Others$ ./superChecker.sh Test
Test is a directory.
sokmeak@sokmeaksaren:~/bin/Others$ ./superChecker.sh getStarted
getStarted is a regular file.
sokmeak@sokmeaksaren:~/bin/Others$ ./superChecker.sh checkFileExtension
checkFileExtension is a regular file.
```

6. Write a shell script that compare two numbers that are accepted as arguments.

```
  GNU nano 7.2                                                    compare.sh
#!/bin/bash

if [ $# -ne 2 ]; then
  echo "Usage: $0 <num1> <num2>"
  exit 1
fi
num1=$1
num2=$2
if [ $num1 -gt $num2 ]; then
  echo "$num1 is greater than $num2."
elif [ $num1 -lt $num2 ]; then
  echo "$num1 is less than $num2."
else
  echo "$num1 is equal to $num2."
fi
```

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod +x compare.sh
sokmeak@sokmeaksaren:~/bin/Others$ ls
checker.sh          compare.sh  eoneb  etwob  etwod          getStarted      mycal.sh    superChecker.sh  user_management
checkFileExtension  eonea       etwoa  etwoc  fileOrganizer  helloworld.jar  organizeMe  Test
```

```
sokmeak@sokmeaksaren:~/bin/Others$ ./compare.sh 3 5
3 is less than 5.
sokmeak@sokmeaksaren:~/bin/Others$ ./compare.sh 5 3
5 is greater than 3.
sokmeak@sokmeaksaren:~/bin/Others$ ./compare.sh 5 5
5 is equal to 5.
sokmeak@sokmeaksaren:~/bin/Others$
```

7. Write a shell script to check to see if the file “file_path” exists. If it does exist, display “file_path passwords are enabled.” Next, check to see if you can write to the file. If you can, display “You have permissions to edit “file_path.””If you cannot, display “You do NOT have permissions to edit “file_path”

```
  GNU nano 7.2                                          check_file_permissions.sh
#!/bin/bash

if [ $# -eq 0 ]; then
  echo "Usage: $0 <file_path>"
  exit 1
fi
file_path=$1
echo " $file_path"
if [ -e "$file_path" ]; then
  echo "$file_path passwords are enabled."
  if [ -w "$file_path" ]; then
    echo "You have permissions to edit \"$file_path\"."
  else
    echo "You do NOT have permissions to edit \"$file_path\"."
  fi
else
  echo "$file_path does not exist."
fi
```

```
sokmeak@sokmeaksaren:~/bin/Others$ chmod +x check_file_permissions.sh
sokmeak@sokmeaksaren:~/bin/Others$ ls
checker.sh          check_file_permissions.sh  eonea  etwoa  etwoc  fileOrganizer  helloworld.jar  organizeMe       Test
checkFileExtension  compare.sh                 eoneb  etwob  etwod  getStarted     mycal.sh        superChecker.sh  user_management
```

```
sokmeak@sokmeaksaren:~/bin/Others$ nano check_file_permissions.sh
sokmeak@sokmeaksaren:~/bin/Others$ ./check_file_permissions.sh ./helloworld.jar
 ./helloworld.jar
./helloworld.jar passwords are enabled.
You have permissions to edit "./helloworld.jar".
sokmeak@sokmeaksaren:~/bin/Others$ ./check_file_permissions.sh /bin
 /bin
/bin passwords are enabled.
You do NOT have permissions to edit "/bin".
sokmeak@sokmeaksaren:~/bin/Others$ ./check_file_permissions.sh /home/sokmeak
 /home/sokmeak
/home/sokmeak passwords are enabled.
You have permissions to edit "/home/sokmeak".
sokmeak@sokmeaksaren:~/bin/Others$ ./check_file_permissions.sh /home/sokmeak/bin/
 /home/sokmeak/bin/
/home/sokmeak/bin/ passwords are enabled.
You have permissions to edit "/home/sokmeak/bin/".
sokmeak@sokmeaksaren:~/bin/Others$
```